Prim’s Algorithm

Prim’s Algorithm-

- Prim’s Algorithm is a famous greedy algorithm.
- It is used for finding the Minimum Spanning Tree (MST) of a given graph.
- To apply Prim’s algorithm, the given graph must be weighted, connected and undirected.

Prim’s Algorithm Implementation-

Step-01:

- Randomly choose any vertex.
- The vertex connecting to the edge having least weight is usually selected.

Step-02:

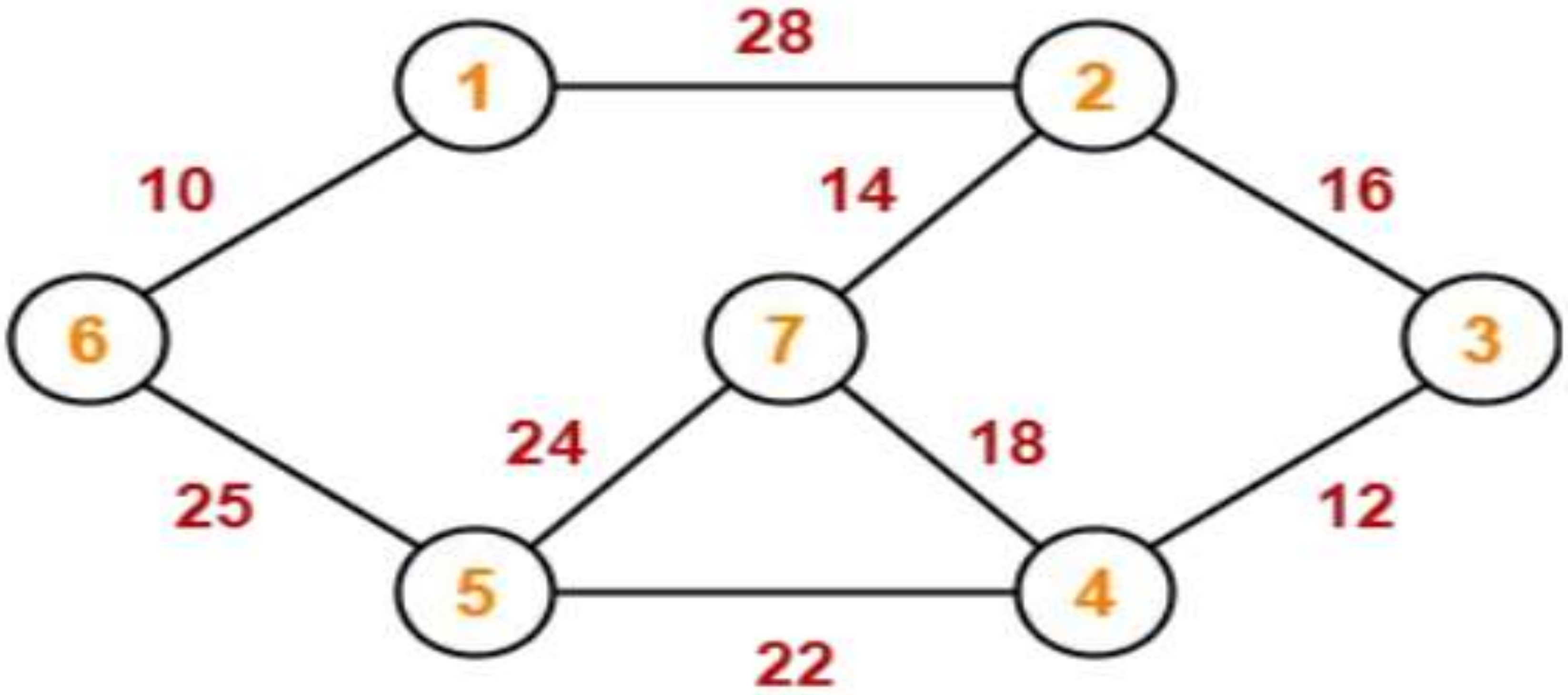
- Find all the edges that connect the tree to new vertices.
- Find the least weight edge among those edges and include it in the existing tree.
- If including that edge creates a cycle, then reject that edge and look for the next least weight edge.

Step-03:

- Keep repeating step-02 until all the vertices are included and Minimum Spanning Tree (MST) is obtained.

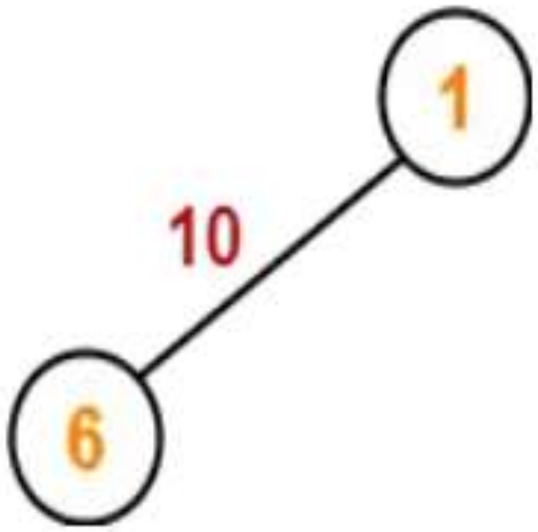
PRACTICE PROBLEMS BASED ON PRIM’S ALGORITHM-

- Construct the minimum spanning tree (MST) for the given graph using Prim’s Algorithm-

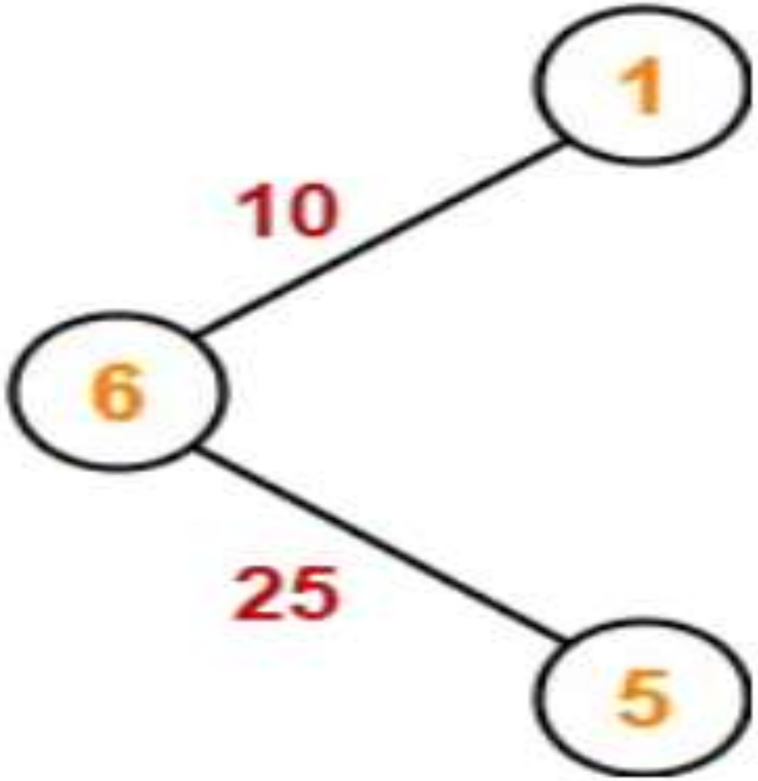


The above discussed steps are followed to find the minimum cost spanning tree using Prim’s Algorithm-

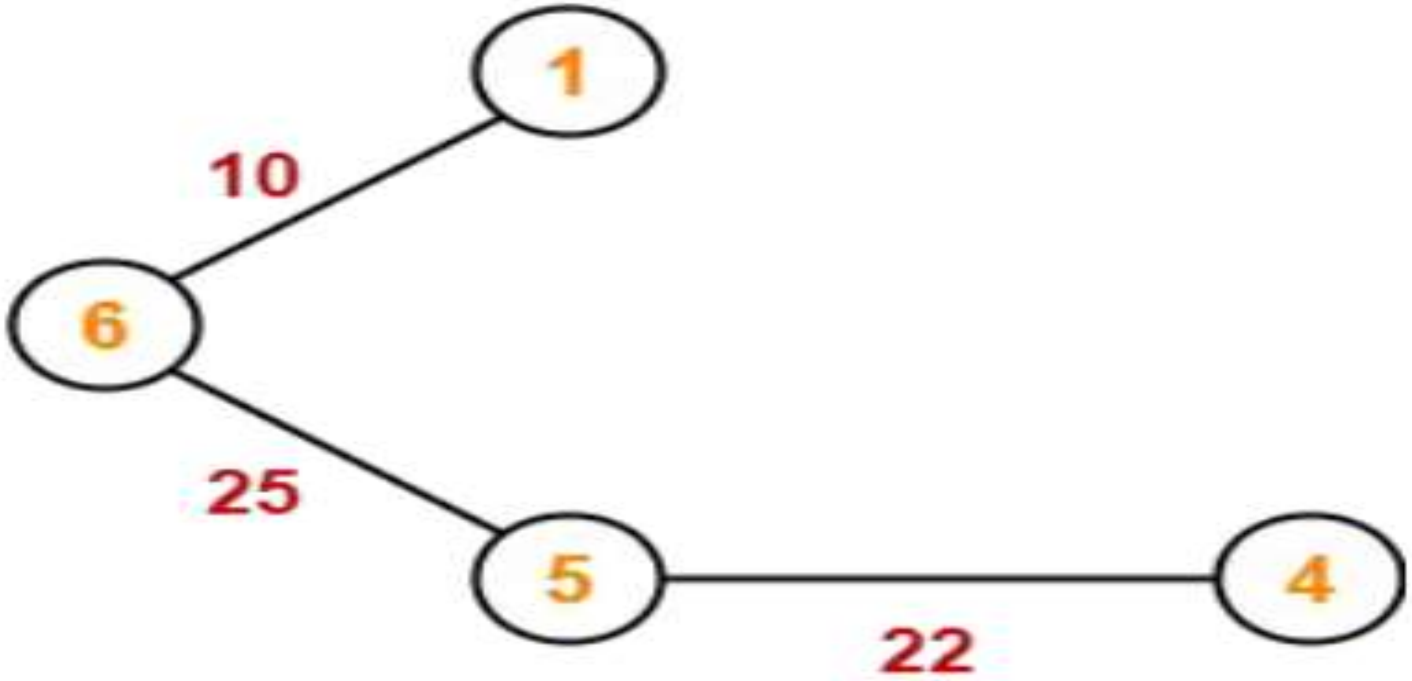
Step-01:



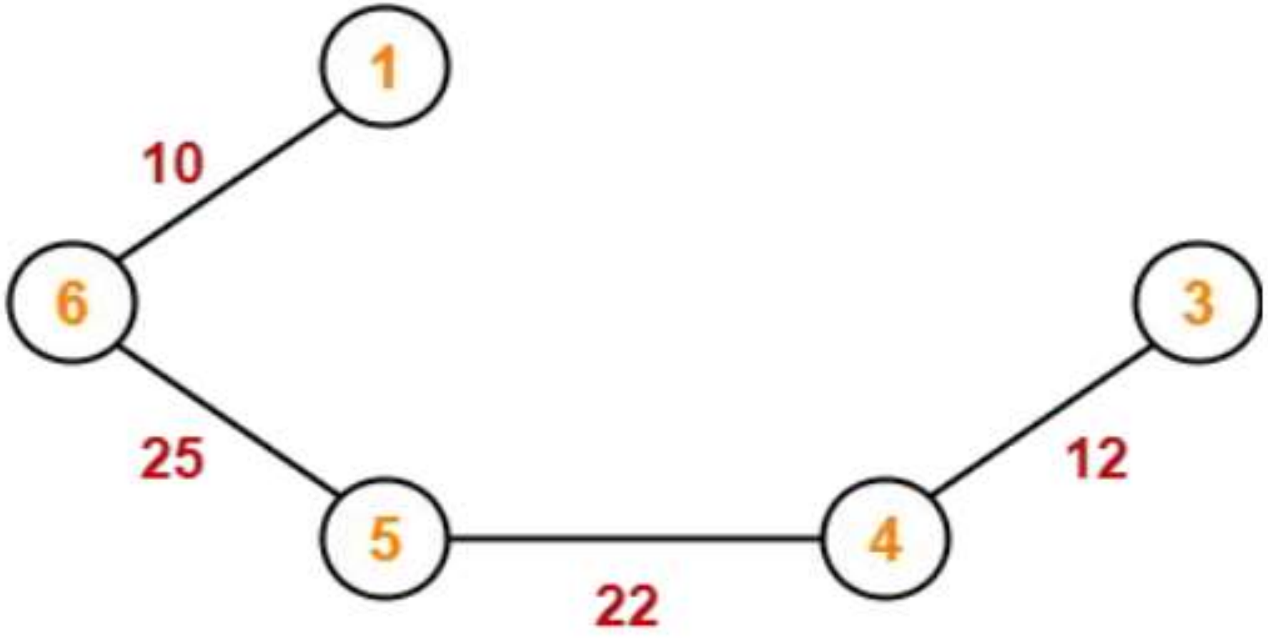
Step-02:



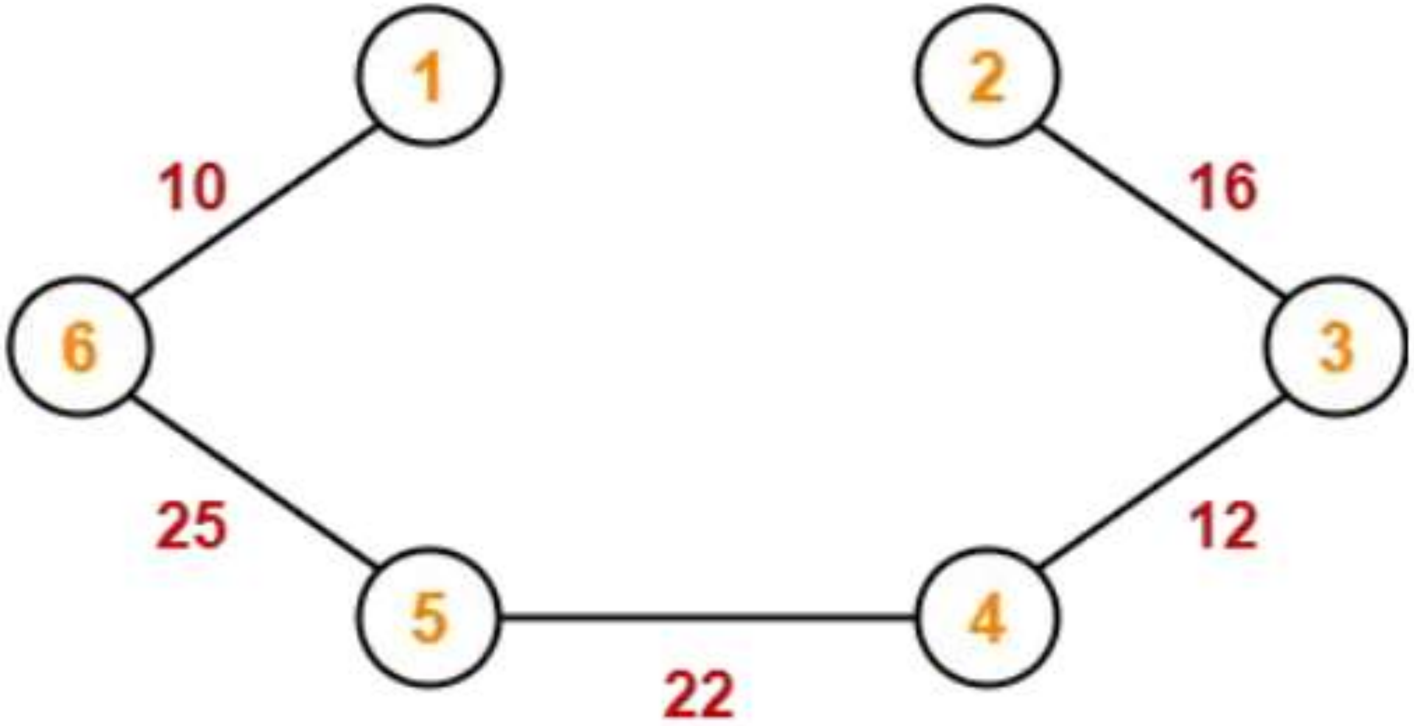
Step-03:



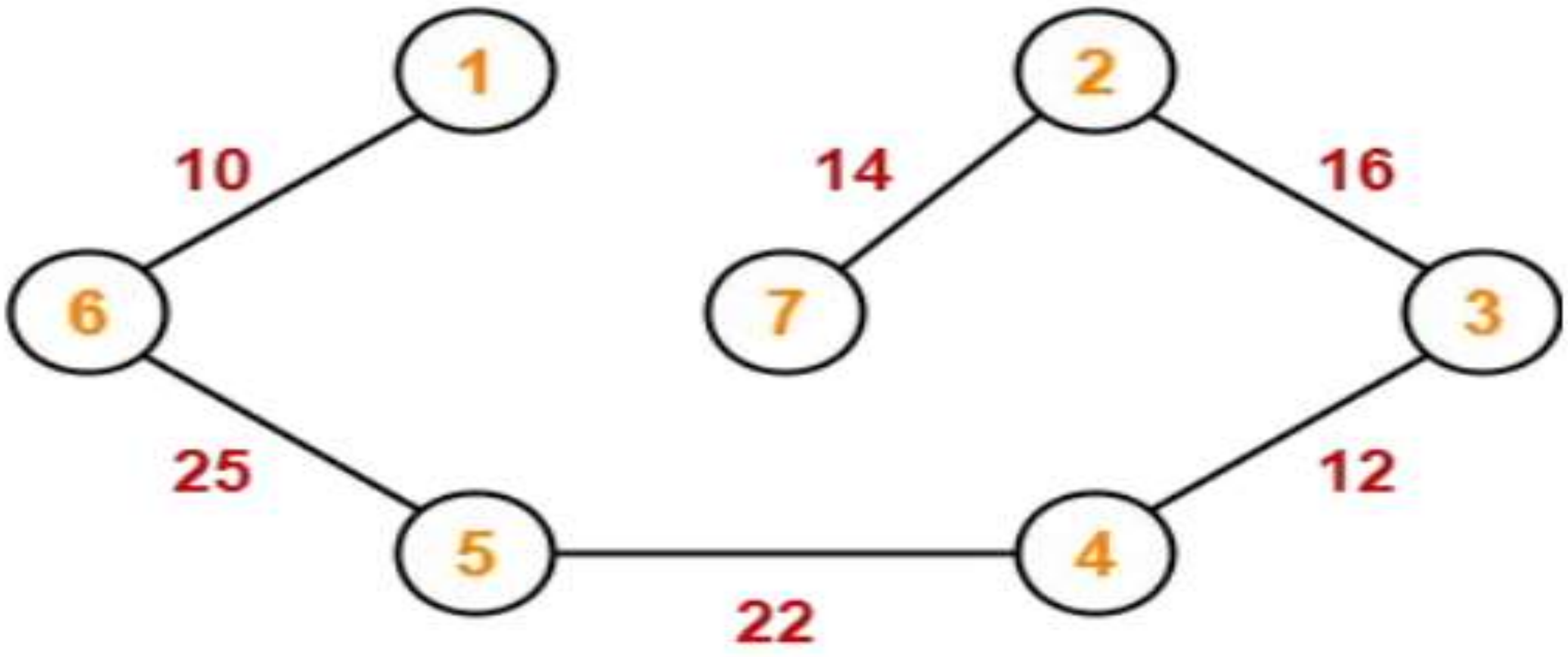
Step-04:



Step-05:



Step-06:



Since all the vertices have been included in the MST, so we stop.

Now, Cost of Minimum Spanning Tree

= Sum of all edge weights

= 10 + 25 + 22 + 12 + 16 + 14

= 99 units